

Machine Learning Exercise 2 Report

Distance Methods & Decision Trees

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1 K-Nearest Neighbors (KNN)

1.1 Task 5.1: Accuracy Analysis

The experiments evaluated KNN performance across varying k values and distance metrics (L_1 , L_2). As shown in Table 1, accuracy degrades significantly as k increases, regardless of the metric used.

Metric	k=1	k=10	k=100	k=1000	k=3000
L1	0.9670	0.9617	0.9231	0.7450	0.4018
L2	0.9680	0.9577	0.9201	0.7417	0.3981

Table 1: KNN Test Accuracy. Performance drops as the neighborhood size grows.

1.2 Task 5.2: Visualization

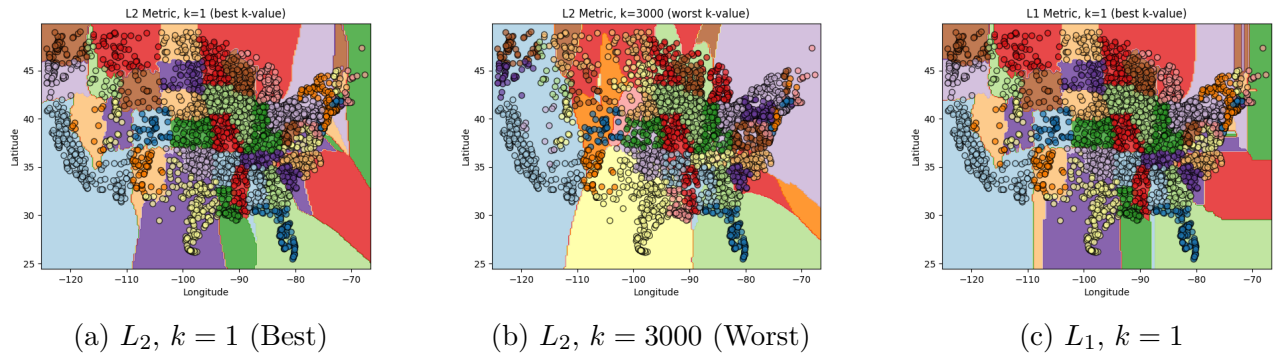


Figure 1: Decision boundaries comparing model complexity and distance metrics.

(a) Effect of k : Bias-Variance Tradeoff

Comparing Figure 1a ($k=1$) and Figure 1b ($k=3000$) illustrates the bias-variance tradeoff. The $k=1$ model is highly flexible (low bias, high variance), allowing it to fit the sharp, irregular boundaries of the US states precisely. Conversely, $k=3000$ forces the model to average votes over a vast area. This introduces high bias (underfitting), effectively smoothing out local geographic details and merging smaller states into larger, indistinct regions. The drop in accuracy from $\sim 96\%$ to $\sim 40\%$ confirms that the global average is a poor predictor for local geography.

(b) Effect of Metric: Geometry of L_1 vs L_2

The choice of metric fundamentally alters the geometry of the decision boundaries. Figure 1a uses the L_2 (Euclidean) metric, where equidistant points form circles. This results in natural, curved boundaries that resemble Voronoi cells. Figure 1c uses the L_1 (Manhattan) metric, where equidistant points form diamonds (rotated squares). Consequently, the decision boundaries tend to align with 45-degree angles or coordinate axes, giving the map a blockier, geometric appearance.

1.3 Task 5.3: Anomaly Detection

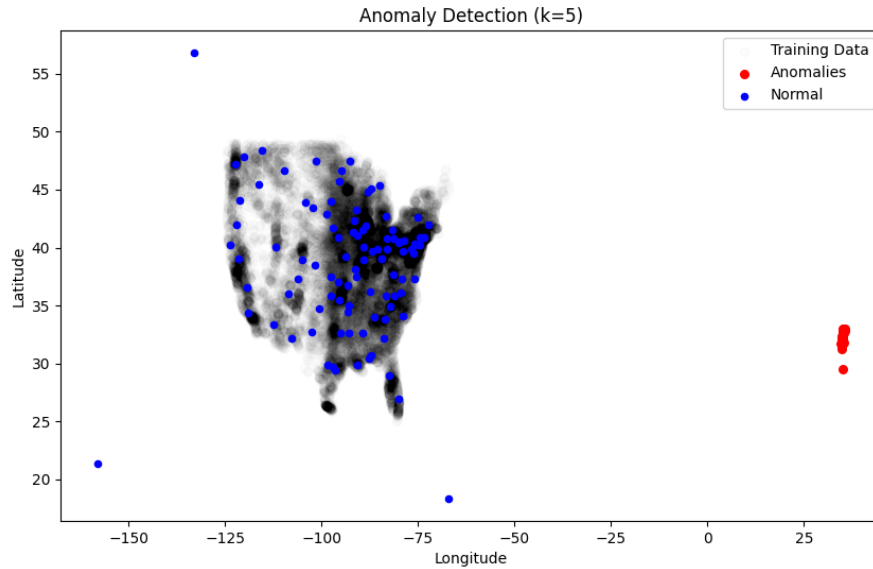


Figure 2: Anomaly Detection: Normal points (Blue) vs Anomalies (Red).

Q 5.4: The results in Figure 2 validate the distance-based anomaly detection approach. Normal points (blue) cluster tightly within the continental US, where the distance to the nearest training neighbors is minimal. Anomalies (red) appear exclusively in data-sparse regions—such as the oceans or across national borders—because the distance to the nearest US city is large, exceeding the anomaly threshold.

2 Decision Trees

2.1 Task 6.1: Hyperparameter Tuning

Depth	Leaves	Train	Val	Test	Depth	Leaves	Train	Val	Test
1	50	0.11	0.11	0.11	10	1000	0.94	0.93	0.92
...	...	...	...	...	20	50	0.86	0.84	0.84
6	50	0.59	0.58	0.58	20	100	0.95	0.94	0.93
6	1000	0.59	0.58	0.58	20	1000	1.00	0.98	0.98
10	50	0.85	0.84	0.83	100	1000	1.00	0.98	0.98

Table 2: Decision Tree performance. Accuracy plateaus around Depth 20.

2.2 Task 6.2: Analysis

(1) Best Tree: The optimal model found via validation was **Depth=20, Leaves=1000**. Its Test Accuracy is **0.9774**.

(2) Generalization: The nearly identical Validation (0.9800) and Test (0.9774) scores indicate the model generalizes well. The validation set was sufficient to select a robust model without overfitting.

(3) Leaf Capacity: While there are 50 states, 50 leaves are insufficient for high accuracy (maxing out at ~ 0.84). Decision trees partition space into axis-aligned rectangles. Complex, non-rectangular shapes (like state borders) require a composite of many small rectangles to approximate accurately. Thus, a 50-leaf limit forces crude approximations, preventing the model from resolving the geometry of all 50 states.

Visualizations (Q4 - Q6)

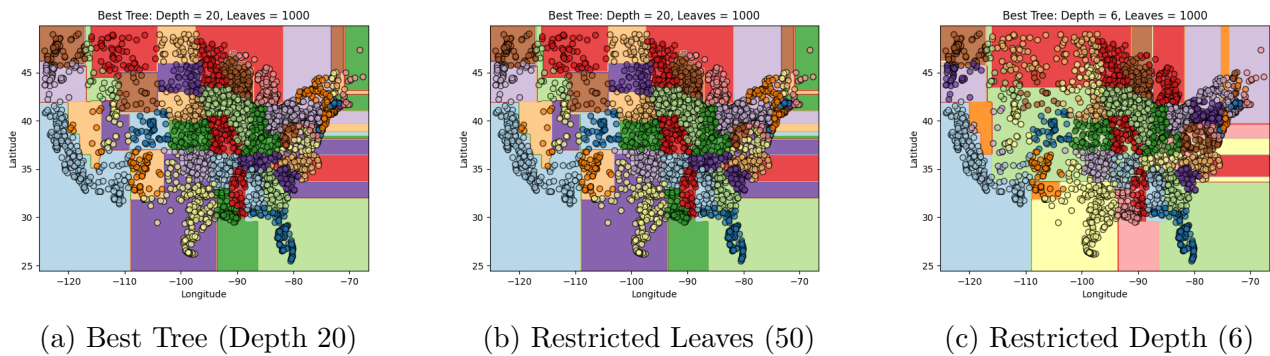


Figure 3: Impact of constraints on Decision Tree boundaries.

(4) Shape of Classes: Figure 3a shows that classes are defined by jagged, rectilinear shapes. Since decision trees split on one feature at a time ($x > \theta$ or $y > \theta$), they are incapable of producing diagonal or curved lines. All boundaries are effectively constructed from "pixels" or rectangular blocks.

(5) Restricted Leaves (Fig 3b): Limiting the model to 50 leaves results in a blocky, low-resolution map. Without enough leaves to "draw" the fine details, the model simplifies complex borders into straight lines or merges neighboring regions, causing the observed drop in accuracy.

(6) Restricted Depth (Fig 3c): Restricting depth to 6 severely limits the model's expressiveness. With only 6 splits allowed from root to leaf, the tree cannot isolate small or specific regions. This high bias results in broad, coarse segmentation where large groups of states are merged into single classes.

2.3 Task 6.7: Random Forests

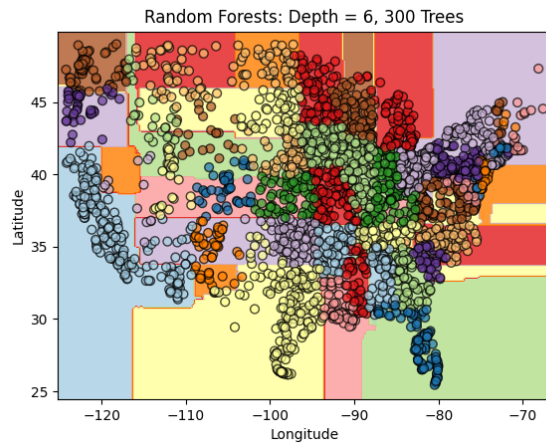


Figure 4: Random Forest (300 Trees, Depth 6)

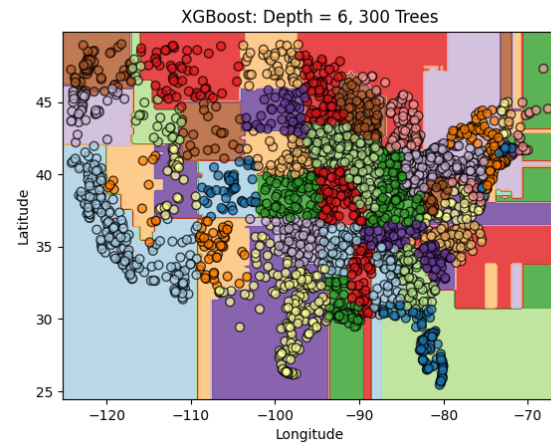


Figure 5: XGBoost (300 Trees, Depth 6)

Results: Test Accuracy = **0.8046**.

Comparison to Q1: The Random Forest is **less expressive** than the single Best Tree (0.80 vs 0.98). While aggregating 300 trees reduced variance and improved upon the single Depth-6 tree (0.58), the ensemble could not overcome the bias inherent in the Depth-6 constraint. The shallow trees simply lack the complexity to capture fine details, regardless of how many are averaged.

2.4 Task 6.8: XGBoost (Bonus)

Results: Test Accuracy = **0.9650**.

Analysis: XGBoost significantly outperforms the Random Forest, nearly matching the Best Tree's performance. **Explanation:** This highlights the advantage of **Boosting** over Bagging for high-bias learners. Random Forest (Bagging) trains independent trees; averaging many "stupid" (shallow) trees yields a smoother, but still "stupid," result. XGBoost (Boosting) trains sequentially, with each new tree correcting the errors of the previous ones. This iterative correction reduces bias, allowing the ensemble to learn complex, sharp boundaries (Fig 5) despite using weak, shallow base learners.

AI Tool Usage

Per the submission guidelines, I acknowledge the use of Gemini 2.5 Pro for the following purposes during this exercise:

- **Coding Assistance:** Generating syntax for data loading with Pandas and classifier initialization with Scikit-Learn/XGBoost.
- **Conceptual Clarification:** Refining my understanding of the Bias-Variance tradeoff and the geometric differences between L_1 and L_2 norms.

- **Report Formatting:** Assisting with LaTeX code organization and table formatting to ensure a professional layout.